F005 | v0.4.0 audit and v0.4.1 repair

19 October 2026 | Development audit, not physical validation

1. Grounded diagnosis

The submitted v0.4.0 run did not calculate any free-run metrics. The waveform-family candidate generator crossed its hard cap **before scoring**, and the code then blocked the independent validation forecast.

Gate / observation	User-run value	Interpretation
Trajectories examined	9,822	Input inventory read
P5b candidate pairs	150,001 / 150,000 max	Hard cap exceeded
P5b scored	0	No near-match conclusions
Large fingerprint buckets	8 skipped	Review incomplete
P6a	BLOCKED	No forecast errors
Holdout outputs	Not accessed	Separation retained

2. Minimal v0.4.1 patch

The patch performs a complete SHA-256 scan of exact input-signal duplicates, then performs a reproducible, capped **approximate** morphology screen. It samples dense buckets rather than skipping them without review. The approximate search is never reported as exhaustive.

Crucially, **P5b remains REVIEW**, while P6a is allowed to run on STANDARD_VALIDATION only, explicitly labeled **DEVELOPMENT_CONDITIONAL_FREE_RUN_FAMILY_REVIEW_OPEN**. No TEST/OOD elevations are used.

3. Reproducible test contract

Frozen ARX(2): A=1.9856661039519918, B=-0.9908696423084401, C=-1.118156917616826; dt=0.004 s. Anchors: 0.004, 1.5, 2.8 s. Horizons: 0.1, 0.5, 1, 2, 3 s. Forcing is a known future exogenous acceleration; elevations only seed the first two points.

Compare each terminal-horizon RMSE against persistence and two-point linear extrapolation. No refit, no test/OOD reveal, no Euler or PKLSA claim.

4. Software verification and remaining gates

Eight synthetic unit and integration tests passed, including holdout-output-access rejection, deterministic bounded candidate screening, detected exact duplicates, and an end-to-end PDF/CSV development run. **The user's 9,822 trajectories were not available in this environment**; v0.4.1 experimental results remain pending local execution.

Family independence requires source-defined family IDs and a group-disjoint holdout. The present fixed candidate partition remains NON-BLIND. P7 (PKLSA) NOT RUN; P8 physical SST conclusions UNRESOLVED.

5. Local invocation

`run_all_v041.cmd` from the existing F005 root; outputs are written to `outputs/f005_free_run_family_v0.4.1/`. Keep existing v0.4.0 folders and source archives unchanged.

Sources: user-uploaded F005 v0.4.0 report/summary; L. Ljung, System Identification: Theory for the User, 2nd ed. (1999); P. G. Saffman, Vortex Dynamics (1992), doi:10.1017/CBO9780511624063.